Satisfiability of first-order sentences is undecidable

The Open Logic Project contributors

openlogic:turing-machines/undecidability:local-snapshot-20261013;
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Editorial note. The counted statement is the author's own corollary environment cor:undecidable-sat (source lines 1537-1539, the \ollabel on line 1537, which is followed by a comment character because the source breaks the line there); the proof environment begins two lines below it and is its own proof, closing at source line 1556.

Editorial note. Neither the statement nor the proof contains an \olref, an \Olref or a \cref, so no source pointer and no bibliography entry are needed; the halting and decision-problem notions used in the proof belong to the retained unit and every symbol is declared in the retained preamble.

Editorial note. The retained preamble is the minimal working preamble shipped with the frozen OLP unit (OLP's own style stack needs prooftree, ntheorem, extarrows, mathdots, isomath and subfiles, which the project toolchain does not have). Author text is untouched; only the packaging is the intake line's.

Editorial note. Section and theorem counters are reset to zero so the retained corollary prints with the numbering of the retained front matter rather than continuing the frozen file's count.

Editorial scope. One original result of the Open Logic Project's undecidability unit, counted once: first-order satisfiability is undecidable.

Allowed standard prerequisites (editorial).

- Turing computability and reductions. A problem is decidable when some Turing machine decides it; if one problem reduces to another by a computable transformation and the target is decidable, so is the source.. The whole proof is the reduction from the decision problem for validity to satisfiability (source lines 1542-1555).
- Validity versus unsatisfiability of the negation. A first-order sentence is valid if and only if its negation is unsatisfiable.. This equivalence is what converts the answer of the assumed machine S into an answer for the decision problem (source lines 1552-1554).
- The decision problem for validity is unsolvable. There is no Turing machine that decides whether an arbitrary first-order sentence is valid.. The derived machine would solve that problem, and the proof closes by contradiction (source lines 1554-1555).

Corollary 0.1. *It is undecidable if an arbitrary !!sentence of first-order logic is satisfiable.*

Proof. Suppose satisfiability were decidable by a Turing machine S. Then we could solve the decision problem as follows: Given !!asentence B as input, move $!B$ to the right one square. Return to square 1 and write the symbol $\neg$.

Now run the Turing machine S. It eventually halts with output either 1 (if $\neg!B$ is satisfiable) or 0 (if $\neg!B$ is unsatisfiable) on the tape. If there is a | on square 1, erase it; if square 1 is empty, write a |, then halt.

This Turing machine always halts, and its output is 1 iff $\neg!B$ is unsatisfiable and 0 otherwise. Since $!B$ is valid iff $\neg!B$ is unsatisfiable, the machine outputs 1 iff $!B$ is valid, and 0 otherwise, i.e., it would solve the decision problem. □